

Big Data, Machine Learning, and AI in Finance

Joanna Wang

Peking University HSBC Business School

Big Data, Machine Learning, and AI in Finance

Course Readings

Core Readings

FSB (2017): “Artificial Intelligence and Machine Learning in Financial Services” — Executive Summary and Sections 1–3

Bachinskiy: “The Growing Impact of AI in Financial Services: Six Examples,” *Medium*

Oliver Wyman: “Artificial Intelligence Applications in Financial Services” — Section on how AI is applied

Regulatory Readings

Mayer Brown (2019): “AI and Big Data Regulatory Risks Under Banking and Consumer Financial Laws”

Stackhouse, St. Louis Fed: “Fintech: Understanding AI Use in Financial Services”

Key Lesson

AI and ML are reshaping credit, trading, compliance, and customer service — but regulatory challenges are mounting equally fast.

Study Questions

Technology and Applications

What are **artificial intelligence**, **machine learning**, and **deep learning**? How do they relate to pattern recognition and decision-making in finance?

How do **NLP**, chatbots, and voice assistants transform the customer interface? What is the future of conversational AI in banking?

Which financial sectors have seen the most AI adoption — credit, trading, compliance, or wealth management?

Policy and Competition

How has AI affected the competitive landscape between **incumbents**, **big tech**, and **fintech startups**?

What public policy challenges does AI present for **fairness**, **explainability**, **privacy**, and **financial stability**?

How do black-box credit models interact with fair lending laws such as ECOA and the Fair Housing Act?

What Are AI, Big Data, and Machine Learning?

Defining the Core Concepts

Artificial Intelligence

AI: “the application of computational tools to address tasks traditionally requiring human sophistication” (FSB, 2017)

Encompasses rule-based expert systems (1970s–1980s), statistical ML (1980s–2000s), and modern deep learning (2010s–present)

Narrow AI (task-specific) dominates today; **general AI** remains aspirational

Big Data

5 Vs: Volume, Velocity, Variety, Veracity, Value

By 2024, over 120 zettabytes created annually; projected to reach 400 ZB by 2028

Financial data: CRSP, Bloomberg, SEC EDGAR, satellite imagery, credit card transactions

Machine Learning

ML: “a method of designing algorithms that optimise automatically through experience and with limited human intervention” (FSB, 2017)

Three paradigms: **supervised** (labeled data), **unsupervised** (unlabeled), **reinforcement learning** (reward signals)

ML powers credit scoring, fraud detection, algorithmic trading, and robo-advisory

Key Lesson

AI, Big Data, and ML are not synonyms — AI is the broadest goal, ML is the primary technique, and Big Data provides the fuel.

The AI Hierarchy: From AI to Deep Learning

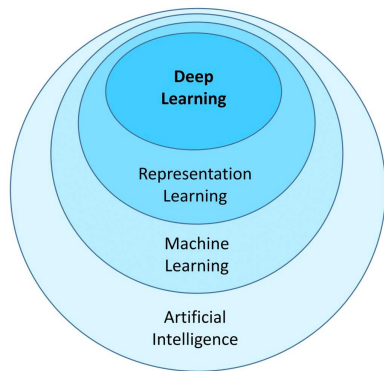
AI (1950s): computers mimic human intelligence; Turing's "Imitation Game" (1950) sets the conceptual foundation

Machine Learning (1980s): machines improve with experience; moves beyond hand-coded rules to statistical pattern recognition

Deep Learning (2010s): multi-layer neural networks; powered by GPUs and massive datasets; AlexNet (2012) sparked the modern era

Large Language Models (2017–present): transformer architecture ("Attention Is All You Need", Vaswani et al.); GPT-4, Claude, Llama 3

Image by Lex Fridman. Used with permission.



Deep Learning Architecture Overview

Deep Learning in One Slide

- **What is it:**
Extract useful patterns from data.
 - **How:**
Neural network + optimization
 - **How (Practical):**
Python + TensorFlow & friends
 - **Hard Part:**
Good Questions + Good Data
 - **Why now:**
Data, hardware, community, tools, investment
 - **Where do we stand?**
Most big questions of intelligence have not been answered nor properly formulated
- **Exciting progress:**
 - Face recognition
 - Image classification
 - Speech recognition
 - Text-to-speech generation
 - Handwriting transcription
 - Machine translation
 - Medical diagnosis
 - Cars: drivable area, lane keeping
 - Digital assistants
 - Ads, search, social recommendations
 - Game playing with deep RL



For the full list of references visit:
<https://deeplearning.mit.edu/references>

<https://deeplearning.mit.edu> 2019

Source: "Deep Learning Basics," Lex Fridman. Used with permission.

Alternative Data and Big Data in Finance

Alternative Data: The New Edge for Investors

Traditional Data Sources

CRSP, Compustat: stock prices, accounting financials

Bloomberg, Reuters: market data, news

SEC EDGAR: 10-K, 10-Q, 8-K filings

Analyst reports, earnings call transcripts

Alternative Data Sources

Satellite imagery: parking lot counts predict retail sales before earnings (Orbital Insight, RS Metrics)

Credit card transactions: Second Measure tracks spending by merchant in real time

Job postings: LinkedIn, Burning Glass reveal hiring plans and tech priorities

Geolocation data: foot traffic to stores, airports, and hospitals

Who Uses Alternative Data?

Hedge funds: Two Sigma, Renaissance, Citadel purchase 100+ alt-data sets per year

Asset managers: BlackRock's **Aladdin** monitors \$21T in assets using alt-data risk models

Banks: JPMorgan COiN reviewed 12,000 credit agreements in seconds (vs. 360,000 lawyer-hours)

Key Lesson

Alternative data gives sophisticated investors an information edge measured in *days* before traditional financial data arrives.

Alternative Data FinTech Landscape

Web traffic data: Similarweb, SemRush measure user engagement before quarterly revenues are reported

Social media sentiment: Twitter/X, Reddit (WallStreetBets), LinkedIn signals used by quantitative funds

App usage: Sensor Tower, Apptopia track daily active users; correlated with subscription revenue

Email receipt data: aggregated from consumer inboxes (with consent) to track purchases at Amazon, Uber, DoorDash

Source: "Alternative Data Goes Mainstream in Financial Services," Datanami (2019).

Alternative Data Sources



AI and Machine Learning in Financial Services

Natural Language Processing in Finance

Core NLP Tasks

Sentiment analysis: Li et al. (2008) showed earnings call tone predicts future earnings surprises and abnormal returns

Named entity recognition: extract company names, amounts, and dates from 10-K filings at scale

Topic modeling: LDA identifies risk topics in SEC filings; Hassan et al. (2019) measure political risk from earnings calls

Text similarity: compare current and prior period 10-K disclosures to detect material changes (Cohen et al., 2020)

Finance Applications

Customer service: chatbots handle routine queries (balance, transactions, FAQs) freeing human agents for complex cases

Compliance: NLP screens millions of trade communications for market manipulation and insider trading signals

Regulatory filing extraction: JPMorgan COiN uses NLP to parse loan documents;

BloombergGPT (2023) trained on 363 billion financial tokens

Key Lesson

NLP converts unstructured text — the majority of financial information — into structured signals for investment and risk decisions.

Bank of America Erica: AI-Powered Banking

Erica launched in 2018; by 2023 surpassed **1 billion client interactions** — the first bank AI to reach this milestone

Handles balance inquiries, transaction searches, bill reminders, credit score updates, and proactive financial insights

Uses **NLP and predictive analytics** to personalize recommendations; learns from 67 million Bank of America clients

Cost per interaction far below human agent cost; reduces call center volume by millions of contacts per year

Source: Bank of America. All rights reserved.



Hi, I am Erica.
See what I can do for you.



AI and Machine Learning — Finance Applications

AI as a Tool (Used Internally)

Credit scoring: Upstart uses 1,600+ variables vs. FICO's 22; claims 27% fewer defaults at the same approval rate

Fraud detection: American Express reduced fraud losses by 60% using real-time ML on transaction streams

Algorithmic trading: Renaissance Technologies Medallion Fund returned 66% annually (before fees) from 1988–2018 using quantitative ML models

Robo-advisory: Betterment and Wealthfront collectively manage \$50+ billion in AUM using automated portfolio optimization

AI as a Service (External Vendors)

Alphasense (2011): AI-powered financial search engine for earnings calls and filings

ComplyAdvantage (2014): anti-money laundering and KYC software; screens billions of transactions

Zest AI (2009): ML credit underwriting; serves community banks and credit unions

Featurespace (2008): real-time behavioral fraud detection using adaptive Bayesian methods

Dataminr (2009): converts real-time Twitter and news into market-moving signals for hedge funds

Big Data in Financial Markets

Platform Solutions Customers Resources About Us Pricing

Start my free trial

CONTENT

Broker & Independent Research
Real-time and aftermarket research from more than 1,500 providers

Expert Transcript Library
Proprietary database of more than 45,000 expert interviews across industries

Company Documents
Presentations, earnings, filings, and more for 68,000+ companies worldwide

News
Top-tier newspapers, trade journals, and other media sources

Regulatory
Regulatory updates from leading global health

FEATURES

Search
AI-fueled smart chat Assistant and Smart Synonyms™ surface relevant insights faster

Discover
GenAI-powered Smart Summaries that surface instant insights from premium content

Monitor
Customizable dashboards and alerts to easily track market trends and competitors

Collaborate
Annotation, highlighting, and note-taking tools to capture and share key insights

Analyze
Table tools that let you model, export, and



ENTERPRISE INTELLIGENCE **NEW**

Generative AI-powered market intelligence platform that unlocks value from your proprietary internal content, delivering an unmatched, tailored picture of any market

EXPERT INSIGHTS

Exclusive access to primary research

Big Data in Banking and Credit

**COMPLY
ADVANTAGE**

Solutions ^ Industries v Insights v Company v

ComplyAdvantage Mesh

An industry-leading and trusted SaaS-based risk intelligence platform that unites global intelligence to beat financial crime.

Customer Screening

Streamline onboarding with advanced AI to swiftly detect and assess risk from individual clients.

Company Screening

Enhance your risk management by quickly evaluating the risk profiles of company entities.

Ongoing Monitoring

Continuously track changes to customers and companies to alert you to any changes in risk status.

Transaction Monitoring

Monitor transactions for AML risk using rules & ML algorithms, ID clustering & graph analysis.

Payment Screening

Boost sanctions compliance, reduce false positives and improve efficiency.

Fraud Detection

Monitor transactions & events in real-time for fraud using out-of-the-box rules & advanced AI.

Fincrim Risk Intelligence

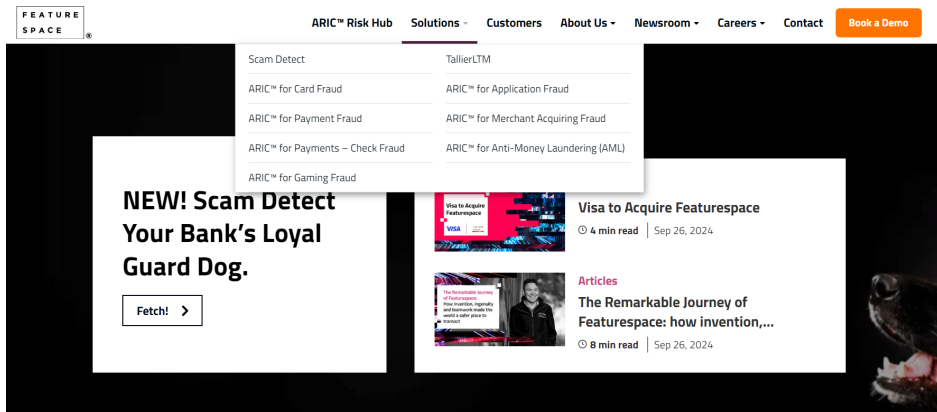
Market-leading global risk and compliance intelligence with extensive coverage.

Sanctions & Watchlists

PEPs & RCAs

Adverse Media

Big Data in Insurance and Risk



FEATURE SPACE

ARIC™ Risk Hub Solutions Customers About Us Newsroom Careers Contact [Book a Demo](#)

Scam Detect
TallierLTM

ARIC™ for Card Fraud
ARIC™ for Application Fraud

ARIC™ for Payment Fraud
ARIC™ for Merchant Acquiring Fraud

ARIC™ for Payments – Check Fraud
ARIC™ for Anti-Money Laundering (AML)

ARIC™ for Gaming Fraud

NEW! Scam Detect Your Bank's Loyal Guard Dog.

[Fetch! >](#)

Visa to Acquire Featurespace

4 min read | Sep 26, 2024

Articles

The Remarkable Journey of Featurespace: how invention,...

8 min read | Sep 26, 2024

AI in Wealth Management and Trading

Tomorrow's technology today

Unlocking AI's vast potential for real-world impact.

We're one of the few companies applying AI at global scale. What makes us different?

With rapid advances in computer vision and deep learning, AI research is on the brink of unlocking AI's tremendous potential. But so far, real-world applications of that potential have been thin on the ground. According to MIT Technology Review, 40% of AI companies don't actually have operational AI to apply. But Tractable's history and growth has been built on applied research from the ground up, and our AI team works closely with industry experts in repairs and insurance to forge real solutions to real problems. For us, AI isn't a dream of the future, but a force to harness right now.

How does it work?

1. Trained on millions of detail-rich images, our AI can undertake a pixel-by-pixel assessment of a photo of a car or home. The AI can then classify the condition or the amount of damage, and provide a detailed estimate.
2. All estimates include a certainty score, depending on part visibility, photo conditions, damage severity and more. The claim can be directed to a human to review, to maintain the highest levels of accuracy. In other cases the estimate can continue through the claims process and speed up payouts.
3. The transformation doesn't stop at customer photos. Our AI can also identify parts that can be salvaged from old or junked cars, for example, and help repair shops source the exact parts they need – improving efficiency and recycling across the whole industries.

Credit Karma and Plaid: Data Aggregation at Scale

Credit Karma (2007)

Free credit report platform monetized through product recommendations; acquired by Intuit for \$7.1 billion in 2020

106 million members; 37 million monthly active users; 2,600+ data points per member

8 billion daily model predictions for personalized financial product matching

Demonstrates that **consumer data at scale** is a standalone business asset

Plaid (2013)

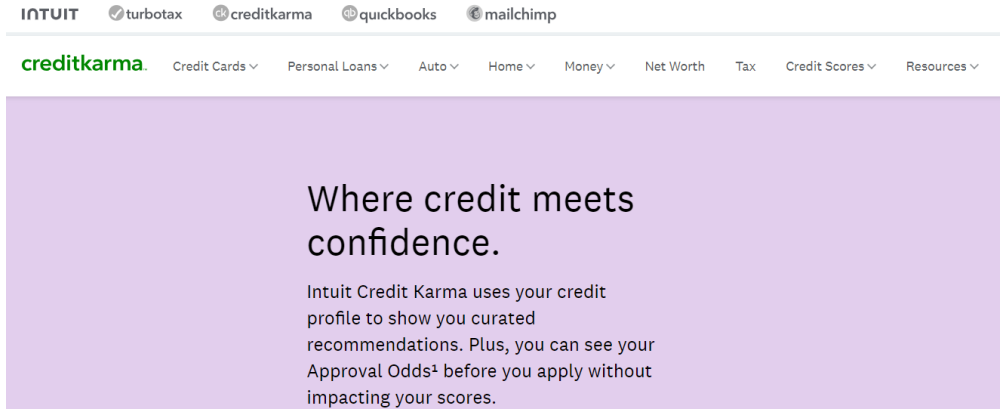
Open API and data aggregator connecting fintech apps to bank accounts; acquired by Visa for \$5.3 billion (2020, later blocked by DOJ)

200+ million linked accounts; serves 11,000 financial companies and 2,600 fintech developers

Powers Venmo, Robinhood, Chime, and hundreds of other apps

Illustrates **open banking infrastructure** as critical fintech plumbing

AI Use Cases Across Financial Services



AI Adoption and Investment Trends



Products ^

Use cases v

For developers v

About us v

Products

Learn how you can
make the most of
financial data

Global coverage >

Pricing >



Payment Initiation >

Instant bank payments
within your app



Balance >

Real-time balance checks



Identity >

Bank account-holder
information



Transactions >

5+ years of categorised
transaction data



Assets >

Point-in-time snapshots of
users' finances



Auth >

Account number, sort
code, IBAN, and BIC



Plaid Link >

Enable your users to
instantly connect their
financial accounts

AI Policy and Regulatory Challenges

Finance Public Policy Framework

Money and Lending: maintaining credit availability and monetary stability; AI changes underwriting at speed and scale

Financial Stability: preventing systemic risk from correlated AI models and algorithmic herding

Consumer and Investor Protection: ensuring fair, transparent, and explainable AI decisions

Anti-Illicit Activity: AML, KYC, and sanctions screening enhanced by ML but also exploitable by adversaries



Key AI Policy Challenges in Finance

Fairness and Explainability

Algorithmic bias: ML credit models trained on historical data may encode racial, gender, or ZIP-code discrimination; CFPB has flagged black-box credit models as a fair lending risk

Explainability (XAI): EU AI Act (2024) and U.S. SR 11-7 require model documentation; “right to explanation” for adverse credit actions under ECOA

Disparate impact: even facially neutral variables (e.g., social media behavior) can proxy protected characteristics

Stability and Systemic Risk

Model risk: correlated AI strategies could amplify market swings; May 2010 Flash Crash erased \$1 trillion in minutes

Algorithmic collusion: competing AI pricing engines may tacitly coordinate without explicit communication

Resiliency: concentration in a few cloud AI providers (AWS, Azure, Google) creates single points of failure

Key Lesson

The faster AI scales in finance, the more urgently regulators need frameworks for *auditability*, *fairness testing*, and *systemic risk monitoring*.

AI and U.S. Consumer Credit Laws

Equal Credit Opportunity Act (ECOA)

Prohibits **disparate treatment** (intentional) and **disparate impact** (neutral policies with discriminatory effects)

Regulation B requires creditors to give **specific reasons** for adverse actions — a challenge for black-box ML models

CFPB (2022): “complex algorithms” that cannot be explained may violate ECOA adverse action notice requirements

Fair Credit Reporting Act (FCRA)

Governs consumer reporting agencies and use of consumer reports in credit decisions

Consumers can dispute inaccurate data — difficult when AI infers scores from non-traditional data

Fair Housing Act

Prohibits discrimination in mortgage lending by race, color, religion, sex, national origin, familial status, or disability

Applies to AI-driven mortgage underwriting and pricing models

CFPB and FTC

UDAAP: AI-generated personalized pricing or targeted offers may constitute deceptive or abusive practices

FTC’s algorithmic accountability framework (2023) calls for pre-deployment bias audits

AI and Privacy Laws

European Union

GDPR (2018): personal data must be processed lawfully, fairly, and transparently; Article 22 limits fully automated decisions with significant effects

EU AI Act (2024): world's first comprehensive AI law; classifies credit scoring and AML as **high-risk AI** requiring conformity assessments and human oversight

United States Federal

Gramm-Leach-Bliley Act: financial institutions must disclose data sharing practices and allow opt-out

FCRA: limits use of consumer report data; alternative data sourced from non-traditional channels may trigger FCRA coverage

U.S. State Laws

California Consumer Privacy Act (CCPA, 2020): consumers can opt out of data sale; right to know, delete, and correct personal data

Illinois **Biometric Information Privacy Act** restricts use of facial recognition for identity verification in financial services

Key Lesson

Privacy law is fragmenting globally — firms operating internationally must reconcile GDPR's "data minimization" with U.S. firms' appetite for maximum data collection.

AI, Finance, and Geopolitical Regulatory Tensions

Global Fragmentation

Finance is **interconnected globally but regulated locally**: a Chinese AI credit model operating in the U.S. faces overlapping FCRA, ECOA, and CFPB scrutiny

International bodies (OECD, FSB, BIS) issue AI principles at a **high level and non-binding** — no enforcement mechanism

Basel Committee (2023) published a discussion paper on AI in banking but stopped short of hard rules

Competing Models

EU: precautionary regulation first — AI Act, GDPR; risk-tiered framework with mandatory audits for high-risk applications

U.S.: sector-by-sector approach; CFPB, OCC, and Fed each issue separate guidance; no unified federal AI law as of 2024

China: state-directed AI deployment in banking; social credit systems blend financial and behavioral data under government oversight

AI versus Human Intelligence

The Turing Test: Can Machines Think?

The Imitation Game (Turing, 1950)

Alan Turing proposed: if a machine can converse indistinguishably from a human, it is “intelligent”

Evaluator (C) converses with human (A) and machine (B); if C cannot reliably identify the machine, it passes

The test sidesteps consciousness questions by focusing on **behavioral equivalence**

Eugene Goostman (2014)

Chatbot simulating a 13-year-old passed at University of Reading; 30 judges, 300 conversations

GPT-4 (2023) routinely passes; humans can no longer reliably distinguish GPT text from human writing

Beyond the Turing Test

Winograd Schema: tests common-sense reasoning; LLMs now exceed 90% accuracy

Professional benchmarks: GPT-4 top 10% on bar exam (2023); passed USMLE at expert level

Limits: AI lacks genuine understanding, embodied experience, and causal reasoning

Key Lesson

Passing the Turing test is no longer remarkable for frontier AI — the key question is whether AI judgment can be *trusted* in high-stakes financial decisions.

The Dartmouth Conference: Birth of AI (1956)

Dartmouth Summer Research Project on Artificial Intelligence (1956): widely considered the **founding event of AI** as an academic discipline

Organized by **John McCarthy** (then Assistant Professor at Dartmouth), **Marvin Minsky**, Claude Shannon, and Nathaniel Rochester

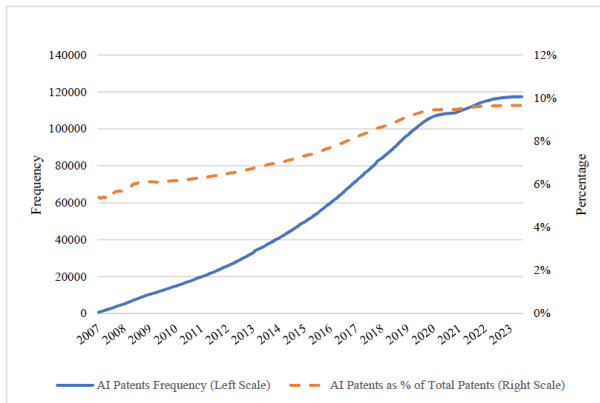
McCarthy coined the term “**Artificial Intelligence**” — deliberately neutral, avoiding loaded terms like “thinking machines”

Goal: clarify whether machines could simulate “every aspect of learning” and “every feature of intelligence”

Dartmouth Summer Research Project on Artificial Intelligence

Date	1956
Duration	Eight weeks
Venue	Dartmouth College , Hanover , New Hampshire
Organised by	John McCarthy , Marvin Minsky , Nathaniel Rochester , and Claude Shannon
Participants	John McCarthy, Marvin Minsky, Nathaniel Rochester, Claude Shannon, and others

AI Displacement or Augmentation? Evidence from Chen and Wang (2025)



Source: Chen and Wang (2025), “Displacement or Augmentation? The Effects of AI on Workforce Dynamics and Firm Value.”

Functional Categories of AI

Perception AI: image recognition, speech recognition, sensor data interpretation (e.g., satellite analysis, computer vision for fraud)

Inference/Prediction AI: credit scoring, risk modeling, algorithmic trading signal generation

Creativity AI: generative content — documents, code, images; GPT-4 for financial report drafting

Engagement AI: chatbots, virtual assistants, recommendation engines (Bank of America Erica, Morgan Stanley AI Assistant)

AI Type (Functional Capability)	Descriptions and Definitions from Psychology, Cognitive Science, and Computer Science	Examples of AI Applications
<i>Language</i>	Phonology and phonotactics, reading, spelling, lexis, morphosyntax, formulaic language, language comprehension, grammaticality, sentence production, and syntax (Carruthers, 2002; Kravchenko, 2007)	Large language models (LLMs); Dubbing for video localization; Voiceprint analysis; Real-time voice translation; Live call-center interpretation
<i>Perception</i>	Reception of stimuli, extraction and storage of perceptual symbols, and updating of perceptual systems (Barsalou, 1999; Wang, Halam, and Hammer, 2022)	Computer vision systems; Autonomous vehicle systems; Diagnostic medical imaging; Augmented reality (AR) and Virtual reality (VR)
<i>Motor control</i>	Control, execution, and adjustment of goal-directed actions based on visual, proprioceptive, and kinesthetic feedback (Grush, 2004; Rosenbaum, 2009)	Autonomous ground vehicles (AGVs), drones, and aircraft; AI-powered humanoid robots; Mobile vehicle swarms; Servo motor control and monitoring; Smart home sensors
<i>Engagement</i>	Selective engagement with the environment based on cognitive motivation and emotion (Simon, 1967; Deci and Ryan, 2000)	Siri; Cortana; Alexa; Chat GPT; customer service AI; Videogame NPCs; Smart home devices; Self-driving delivery robots

Functional Categories of AI: Detailed Map

Builds on parallels between AI capabilities and human cognitive functions (Turing, 1950; Newell, 1960)

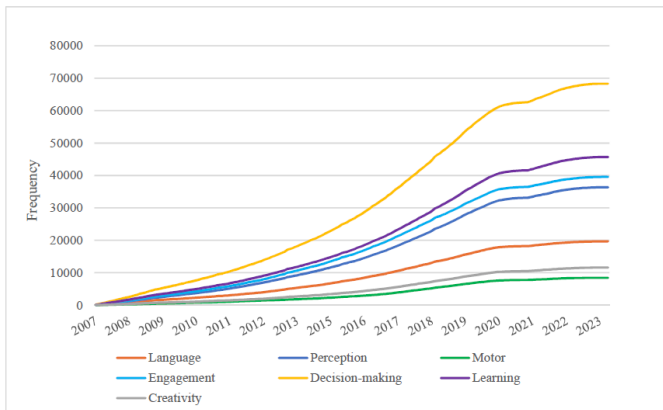
High-level cognitive AI (reasoning, creativity) tends to **augment** skilled workers and increase labor demand

Low-level cognitive AI (routine tasks, data entry, classification) tends to **displace** workers

Finance implication: analysts and portfolio managers augmented; back-office processing and compliance roles automated

<i>Decision-making</i>	Setting plans through a reasoning function and selecting subjectively rational choices based on information about uncertainties and preferences (Pomeroy, 1997; Fox, Cooper, and Glasspool, 2013)	Automated credit decision; AI resume screening; Autonomous vehicle navigation
<i>Learning</i>	Acquiring skills or experience in one situation and utilizing them in another situation (Greeno, Collins, and Resnick, 1996; Silver et al., 2021)	AlphaFold protein structure prediction; Personalized recommendation engines; Machine learning systems for healthcare diagnostics; Fraud and anomaly detection systems
<i>Creativity</i>	Actively and consciously pursuing an innovative solution or a new idea with the novelty derived from imagination (Pelaprat and Cole, 2011; Stokes, 2014)	ChatGPT; Adobe Firefly art generator; OpenAI Sora video generator; Robotic painters and sculptors

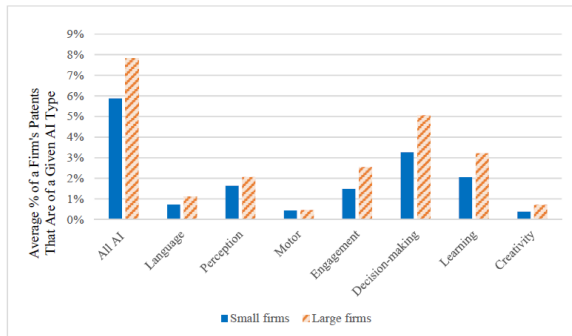
Distribution of AI Patents Across Functional Categories



Source: Chen and Wang (2025).

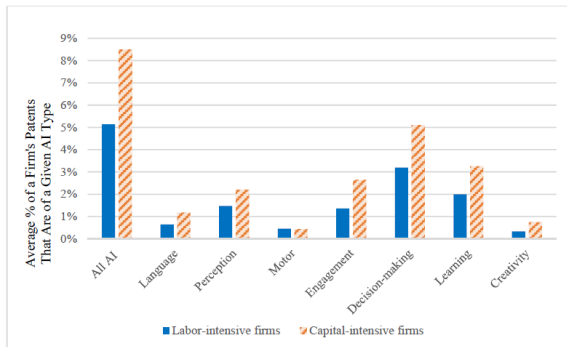
AI Patents by Firm Group

Panel A. AI innovation activity by large and small firms



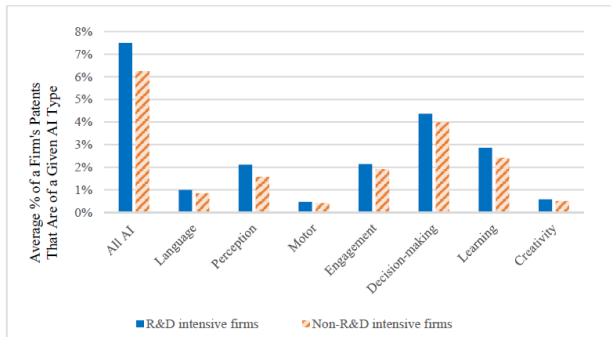
AI Patents by Firm Group: Technology vs. Finance

Panel B. AI innovation activity by labor-intensive and capital-intensive firms



AI Patents: Growth Over Time

Panel C. AI innovation activity by R&D-intensive and non-R&D-intensive firms



AI Patents and Occupational Exposure



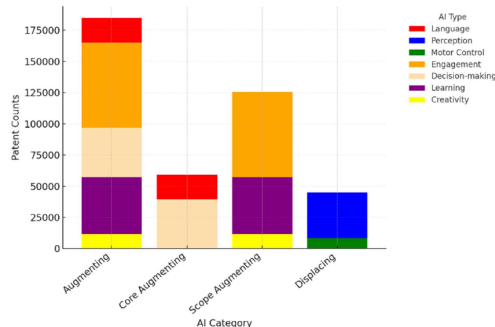
AI Impact on Labor: Augmentation versus Displacement

High-level cognitive AI increases labor demand by expanding the *scope* of what workers can do (scope augmentation)

AI generating innovative or unpredictable outcomes broadens job scope; AI producing predictable outputs tends to displace routine workers

Finance example: ML tools that surface new investment ideas augment analysts; automated report generation displaces junior analysts

Evidence: AI patents rose from 2% of all USPTO patents in 2007 to 10% in 2023 (Chen and Wang, 2025)



Research Design: AI Effects on Workforce and Firm Value

Research Question

How does AI technology affect firm-level **employment** and **firm value** (Tobin's Q)?

Key contribution: study AI by **functional capability** (perception, inference, creativity, engagement) rather than treating all AI as homogeneous

Data

USPTO BDSS: AI patents classified by functional category, 2007–2023

Revelio Labs: individual worker roles, hiring, and separation data for matched firms

Identification: variation in patent grant probabilities driven by *exogenous* examiner assignment lotteries

Key Findings

AI innovation grew from 2% to 10% of all patents (2007–2023)

Augmenting AI increases labor demand for skilled workers; **displacing AI** reduces employment for routine-task workers

Both types increase **Tobin's Q** (firm value), but augmenting AI also raises **TFP** (total factor productivity)

Effect on firm value varies with the firm's **labor market bargaining position**

Effects of AI on Employment: Augmentation

Panel B: Instrumental variables regressions (2SLS, 2nd stage)

	Patent Type $k =$						
	Language AI (1)	Perception AI (2)	Motor control AI (3)	Engagement AI (4)	Decision-Making AI (5)	Learning AI (6)	Creativity AI (7)
Exposed $\times$ Log type- k patents	1.296*** (2.83)	-1.397*** (-2.44)	-1.866** (-2.04)	1.640*** (2.84)	1.397** (2.48)	1.979*** (2.73)	1.308** (2.41)
Exposed $\times$ Log other patents	-9.179 (-1.52)	-6.707 (-1.13)	-8.134 (-1.33)	-9.716 (-1.60)	-9.432 (-1.66)	-9.733* (-1.71)	-9.080 (-1.50)
[1-Exposed] $\times$ Log all patents	-8.407 (-1.38)	-8.462 (-1.39)	-8.210 (-1.32)	-8.352 (-1.36)	-8.273 (-1.40)	-8.330 (-1.40)	-7.648 (-1.24)

Dependent variable: growth in labor demand (augmentation)

Augmenting AI patents cause a statistically significant increase in total employment relative to control firms

Effect concentrated in **skilled, high-cognitive workers:** software engineers, data scientists, financial analysts

New job titles emerge that did not exist before the AI patent grant (**scope augmentation**)

Effects of AI on Employment: Scope Augmentation

Panel B: Instrumental variables regressions (2SLS, 2nd stage)

	Patent Type $\hat{\epsilon} =$				
	Language AI (1)	Engagement AI (2)	Decision-Making AI (3)	Learning AI (4)	Creativity AI (5)
Exposed $\times$ Log type-k patents	0.002 (0.88)	0.008*** (3.52)	0.004 (1.63)	0.011*** (3.64)	0.005*** (3.01)
Exposed $\times$ Log other patents	-0.030*** (-3.08)	-0.034*** (-3.38)	-0.031*** (-3.58)	-0.035*** (-3.76)	-0.031*** (-3.11)
[1-Exposed] $\times$ Log all patents	-0.027*** (-2.75)	-0.025** (-2.38)	-0.026*** (-2.84)	-0.025** (-2.48)	-0.020** (-2.02)

Scope augmentation: growth in demand for workers with *new* skills that did not exist before the AI innovation

Strong effect for **creativity and engagement AI** categories; weaker for pure prediction/inference AI

Consistent with **task-based models** of automation: AI expands the task frontier rather than simply replacing existing tasks

Finance analogy: algorithmic trading created *more* quantitative analyst roles, not fewer, in the decade after Renaissance's success

AI and Firm Efficiency: Productivity and Value

Augmenting	Scope- Augmenting	Core- Augmenting	Displacing
2SLS (2 nd Stage)			
(5)	(6)	(7)	(8)
1.324*** (2.786)	1.036** (2.558)	1.162** (2.403)	0.303 (0.811)
Augmenting	Scope- Augmenting	Core- Augmenting	Displacing
2SLS (2 nd Stage)			
(5)	(6)	(7)	(8)
0.347** (2.135)	0.335** (2.418)	0.321** (2.061)	0.414*** (2.944)

Firm productivity (TFP) increases significantly for **augmenting AI** patents but not for displacing AI patents

Firm value (Tobin's Q) increases for *both* augmenting and displacing AI, reflecting investor expectations of future cost savings and efficiency

Key insight: displacing AI raises market value without raising productivity — consistent with **labor cost reduction** as the primary channel

Finance implication: markets reward AI-driven efficiency even when workers lose jobs

Generative AI and LLMs in Finance

Generative AI in Legal and Financial Services

JPMorgan COiN (Contract Intelligence): NLP system reviews 12,000 commercial credit agreements in *seconds* vs. 360,000 annual lawyer-hours previously required

Legal industry: generative AI cuts billable hours for contract review, due diligence, and regulatory filing drafting — FT estimates 30% reduction in entry-level legal work

Compliance automation: AI reviews trade communications, flags potential market abuse, and drafts SAR (Suspicious Activity Report) narratives for regulators

Risk: over-reliance on AI outputs without human review; “hallucination” in legal citations is a documented problem (Mata v. Avianca, 2023)

Law Firm Allen & Overy Debuts AI Contract Negotiation Tool

BY PYMNTS | DECEMBER 21, 2023



Enterprise AI Platforms: Deloitte PairD

PairD (Deloitte AI Institute): internal generative AI platform deployed to Deloitte's 150,000+ professionals for drafting, coding, and research tasks

Designed to work *securely* within Deloitte's firewall — client data never exposed to public LLM providers

Productivity gains: studies show 30–40% reduction in time for routine document tasks when AI assists knowledge workers

Enterprise AI trend: major financial institutions (Goldman Sachs, Morgan Stanley, HSBC) are deploying private LLMs trained on proprietary data

Article • Machine Learning

Deloitte introduces PairD AI Chatbot for its Workforce

By Amber Jackson
January 08, 2024 • 5 mins

Generative AI for Business Operations

Cognizant and Microsoft announce global partnership to expand adoption of generative AI in the enterprise, and drive industry transformation

April 22, 2024 | Microsoft Source

Microsoft-Cognizant partnership: \$1 billion deal to deploy Copilot across financial services clients; targets business transformation at scale

Key use cases: automated earnings call summaries, regulatory filing generation, client onboarding document preparation, investor Q&A bots

Cost economics: large banks spend \$1–4 billion/year on compliance — generative AI could reduce this by 20–30% (McKinsey, 2023)

Judson Althoff (Microsoft): “generative AI will revolutionize client interactions and free up time to serve clients”

AI in Healthcare Finance: Med-PaLM2

Google Med-PaLM2: scored 85%+ on USMLE-style questions — “expert physician level” and well above the 60% passing threshold

Financial parallel: AI scoring at expert level for medical licensing exams foreshadows similar breakthroughs for CFA, CPA, and actuarial exams

Medscape 2023: 56% of U.S. physicians are enthusiastic about AI as a clinical adjunct; 65% concerned about fully autonomous AI diagnosis

Lesson for finance: human oversight remains essential even when AI outperforms humans on structured benchmarks



AI in Healthcare: Med-PaLM 2 Surpasses Human Doctors in Medical Knowledge Assessment



Avisha NessAiver · Follow

3 min read · May 19, 2023

Morgan Stanley AI Assistant: Wealth Management

Morgan Stanley AI @ Morgan Stanley Assistant (OpenAI GPT-4): gives 16,000 financial advisers instant access to the bank's entire research library

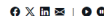
Queries that previously required 30-minute database searches now answered in *seconds*; advisers can spend more time with clients

Co-President Andy Saperstein: “financial advisers will always be the center of our universe — AI frees them to do what they do best”

AUM context: Morgan Stanley manages \$5.5 trillion; even a 1% improvement in adviser productivity is worth billions in retained assets

Morgan Stanley to Launch AI-Powered Assistant for Financial Advisers

BY PYMNTS | SEPTEMBER 18, 2023



AI Creativity: Content Generation at Scale

Canva Magic Studio: converts financial presentations, pitch decks, and investor reports into multi-format outputs (blog, email, social) automatically

Adopted by FedEx, Zoom, Salesforce — demonstrates enterprise-scale creative AI adoption beyond consumer use

Finance application: IR (investor relations) departments use generative AI to produce earnings release drafts, ESG reports, and annual report narratives

McKinsey (2023): generative AI could add \$2.6–4.4 trillion annually to the global economy; financial services is one of the highest-impact sectors

[CREATORS](#) / [TECH](#) / [WEB](#)

Canva's new AI tools automate boring, labor-intensive design tasks

AI Perception: Computer Vision in Finance

Satellite imagery analysis: hedge funds use computer vision to count cars in retail parking lots, ships in ports, and construction equipment at mines — all before official data is released

Insurance claims: Tractable (2014) uses AI image recognition to assess vehicle damage from photos in minutes vs. days for human adjusters; deployed by Geico, Tokio Marine

KYC and fraud: facial recognition and document analysis (passport, ID) in digital onboarding; reduces account opening time from days to minutes

Construction monitoring: AGMIS uses computer vision to enforce safety compliance on construction sites, reducing liability for insurers

agmis

[About Us](#) [Services](#) [Case Studies](#) [News](#) [Career](#) [Contact](#) [EU Projects](#)

Computer vision: AI ensures Worker Safety in Industrial and Construction Sites

Technology company AGMIS launched a Computer vision solution to improve worker safety in high-risk working conditions. The companies currently testing the solution note that it has a lot of real life potential.

AI Engagement: IKEA and the Chatbot Revolution

IKEA (Ingka Group) Billie: AI chatbot handles routine customer service; allows human agents to shift to higher-value home improvement consulting roles

Ingka Group HR: “We are committed to reskilling co-workers through lifelong learning and creating new jobs” — illustrates workforce transition challenge

Financial services parallel: Klarna’s AI assistant (powered by OpenAI) does the work of 700 full-time agents, handling 2.3 million conversations in its first month

Customer satisfaction scores equal to human agents; resolution time cut from 11 minutes to 2 minutes

IKEA Uses AI to Transform Call Center Employees Into Interior Design Advisors

BY PYMNTS | JUNE 14, 2023



Walmart and Enterprise AI Deployment

Walmart deployed generative AI tools to **50,000+ corporate employees** in 2023; one of the largest enterprise AI rollouts globally

Use cases: supplier negotiation preparation, inventory analytics summaries, financial planning document drafting, and benefits administration

Walmart position: AI will **not replace workers** but reduce time on “monotonous, repetitive tasks” — a common framing from large employers

Finance parallel: Goldman Sachs’ internal AI tool (Marcus) automates routine credit analysis tasks; JPMorgan’s LLM-Suite deployed to 60,000 employees

DIVE BRIEF

Walmart rolls out generative AI-powered assistant to 50K employees

The retailer joins a growing group of large U.S. enterprises equipping workers with generative AI capabilities customized to internal systems.

Published Aug. 30, 2023

Klarna: AI Replaces 700 Customer Service Jobs

Klarna's AI bot is doing the work of 700 employees. What will happen to their jobs?

Klarna (buy-now-pay-later leader, 150M users) announced in early 2024 that its AI assistant equaled the work of 700 full-time customer service agents

After its 2021 peak of 5,000 employees, Klarna's headcount was reduced to 3,800; company expects **fewer than 2,300** following AI automation

Hiring freeze: Klarna froze all non-engineering hires; engineers needed to maintain AI systems

Lesson: productivity gains from AI are real and measurable — but so are the employment consequences for service workers

Big Data in Finance: Applications and Ecosystem

The Big Data Revolution in Finance

Scale of the Data Explosion

By 2024, over **120 zettabytes** of digital data created annually; 90% created in the last two years alone

Sources: 15 billion+ IoT sensors, 5 billion smartphone users, 500 million daily tweets, 4 trillion financial transactions per year

Cloud computing makes storage and processing affordable: AWS, Azure, GCP offer petabyte-scale financial analytics at commodity prices

Structured vs. Unstructured Data

Structured: prices, volumes, accounting ratios, credit scores — traditional finance data

Unstructured: text (10-K filings, earnings calls), images (satellite), audio (voice calls) — the new

Financial Data Ecosystem

Market data: CRSP, Bloomberg, Refinitiv, ICE Data Services

Corporate filings: SEC EDGAR (10-K, 10-Q, 8-K, proxy statements, S-1)

Alternative data vendors: Second Measure (credit card), Orbital Insight (satellite), Bombora (web traffic), Envestnet Yodlee (bank transactions)

Research databases: WRDS (Wharton), FactSet, S&P Capital IQ

Key Lesson

The competitive advantage in modern finance increasingly comes from *who has better data* and *who can process it faster* — not just who has better theory.

Textual Data: Analyst Reports

Analyst report NLP: ML models extract tone, specificity, topic emphasis, and forward-looking language beyond standard EPS estimates

Li et al. (2008) showed **earnings call tone** predicts future earnings surprises and cumulative abnormal returns

Soft information (qualitative guidance, management confidence signals) is increasingly processed by ML models used by institutional investors

Huang et al. (2014): readability of MD&A sections is negatively correlated with future earnings and positively correlated with earnings management

Reasons To Buy:

- ▲ Facebook continues to witness significant traction in online and mobile advertising spending. The company intends to capture the opportunity presented by ever-increasing video viewing on social media platforms. Online video is the most lucrative component of digital advertising. As video ads generate more revenues than its photo and text-based substitutes, Facebook is trying to incorporate more and more video-oriented content to bring in more ad dollars. The company launched Watch, a dedicated tab for video viewing, to achieve its goal.
- ▲ Instagram has emerged as an important cash cow for Facebook after introducing its ad platform to worldwide advertisers. To attract more advertisers (over 3 million and counting), Facebook has unveiled tools to promote posts and evaluate business performance directly on Instagram. Moreover, the Facebook is also looking for ways to monetize Instagram Stories, and adding the Checkout feature to the platform is a step toward that direction. The feature allows Instagram users to browse and purchase products from 23 top brands in the United States, all within Facebook's app. Further, solid adoption of the Explore tab, which is used by more than 50% of Instagram users every month, increases the platform's monetization opportunities. Recently, the company started placing ads on the Explore tab that is expected to drive the top line.
- ▲ Messenger and WhatsApp are the other extremely prized possessions. Facebook is aggressively working on monetizing the opportunities presented by its subsidiaries. The company is pretty excited as it opened the Messenger app, which has more than 1 billion users, to developers for creating chatbots that would enable businesses to extend customer service and other transactions. Facebook is rewriting the app from scratch to make it the fastest and most secure major messaging platform in the world. It is also working to bring end-to-end encryption and ephemerality to boost user experience.
- ▲ Ephemerality and end-to-end encryption are also driving popularity of WhatsApp. Per the company, WhatsApp Status is the most popular ephemeral Stories product in the world. Facebook continues to promote WhatsApp business as a private way of interaction between businesses and customers. Rapid adoption of WhatsApp Business by small and mid-sized businesses is noteworthy. The company is building new tools like product catalogs that entrepreneurs can use for free to attract prospective customers. Moreover, increased communication enhances chances of Facebook penetrating the lucrative payments market. The company continues to test the payments feature on WhatsApp in India and intend to launch the same in other countries soon.
- ▲ Oculus is the result of the company's ambitious augmented reality/virtual reality (AR/VR) efforts. AR/VR technology is fast emerging as a lucrative business opportunity. Plus, Facebook said it will be spending over \$3 billion in the next 10 years on VR. The company is investing heavily to develop VR products. Facebook is also investing in VR headset with no wires and full freedom of movement. The company is planning to launch its first generation VR device starting at \$299. The company is also planning to release a pair of Ray-Ban branded smart glasses in 2021. Facebook also unveiled Project Aria that will help it develop the first generation of AR glasses. The company is also planning to release a pair of Ray-Ban branded smart glasses in 2021. Facebook also unveiled Project Aria that will help it develop the first generation of AR glasses.

Facebook benefits from increasing mobile ad revenues, growing adoption of Stories by advertisers across Instagram, core Facebook app and Messenger, and initiatives to improve security.

Textual Data: Patent Text and Firm Innovation

Patent text reveals a firm's technological trajectory *before* financial results materialize — a leading indicator of future revenue and competitive advantage

NLP classifiers identify AI-related, biotech, clean-energy, and defense technology patents at scale across millions of USPTO filings

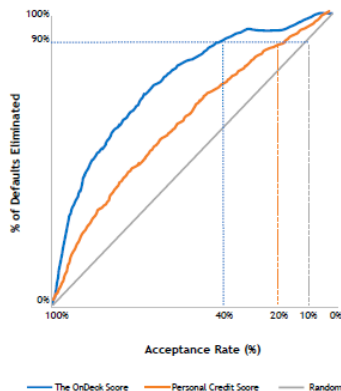
Textual similarity across patent filings (Kelly et al., 2021) captures knowledge spillovers and technology diffusion between firms and industries

Used in Chen and Wang (2025) to classify AI patents by functional category using deep learning fine-tuned on USPTO descriptions

United States Patent [19]		[11] Patent Number:	5,723,765
Oliver et al.		[45] Date of Patent:	Mar. 3, 1998
[54] CONTROL OF PLANT GENE EXPRESSION			
[75] Inventors: Melvin John Oliver, Lubbock; Jerry Edwin Quisenberry, Idaho; Norma Lee Glover Trullinger, Quatah, all of Tex.; Don Lee Kelm, Leland, Miss.			
[73] Assignees: Delta and Pine Land Co., Scott, Miss.; The United States of America as represented by the Secretary of Agriculture, Washington, D.C.			
[21] Appl. No.: 477,559			
[22] Filed: Jan. 7, 1995			
Related U.S. Application Data			
[63] Continuation-in-part of Ser. No. 283,604, Aug. 1, 1994, abandoned.			
[51] Int. Cl.⁶ C12N 15/29; C12N 15/32; A01H 4/00; A01H 5/00			
[52] U.S. Cl. 806/205; 800/250; 536/24.1; 536/23.6; 536/24.5; 435/320.1; 435/240.4; 435/172.3			
[58] Field of Search 536/24.1, 23.6, 536/24.5; 435/320.1, 240.4, 172.3; 800/205, 250			
[56] References Cited			
U.S. PATENT DOCUMENTS		(List continued on next page.)	
4,999,317	9/1990 Sauer	435/172.3	Primary Examiner —Douglas W. Robinson Assistant Examiner —Thomas Hans Attorney, Agent, or Firm —Rothwell, Figg, Ernst & Kurz [57] ABSTRACT A method for making a genetically modified plant comprising regenerating a whole plant from a plant cell that has been transfected with DNA sequences comprising a first gene
5,139,135	10/1992 Umbeck	800/205	
5,217,889	6/1993 Steinman et al.	435/172.3	
5,231,019	7/1993 Paszkowski et al.	435/172.3	
5,244,802	9/1993 Rangan	435/240.5	
5,270,201	12/1993 Richards et al.	435/240.4	
5,304,730	4/1994 Lawson et al.	800/205	
5,332,609	10/1994 Staley et al.	435/240.4	
Foreign Patent Documents		Other Publications	
1,100,000		Lasser, Michael and Bujard, Hermann, "Promoters largely determine the efficiency of repressor action," <i>Proc. Natl. Acad. Sci. USA</i> , 85: 8973-8977 (1988).	
1,100,000		Araki, Kimi et al., <i>Site-specific recombination of a transgene in fertilized eggs by transient expression of Cre recombinase</i> , <i>Proc. Natl. Acad. Sci. USA</i> , 92(1):160-164 (1995).	
1,100,000		Medberry, Scott L., et al., <i>Intra-chromosomal rearrangements generated by Cre-lox site-specific recombination</i> , <i>Nucleic Acids Research</i> , 23(3):485-490 (1995).	
1,100,000		Chapman, Sean, Kavanaugh, Tony and Baulcombe, David, "Potato virus X as a vector for gene expression in plants," <i>The Plant Journal</i> , 2(4):549-557 (1992).	
1,100,000		Odell, Joan T., et al., <i>Seed-Specific Gene Activation Mediated by the Cre/Lox Site-Specific Recombination System</i> , <i>Plant Physiol.</i> , 106:447-458 (Oct. 2, 1994).	
1,100,000		Qin, Minmin, et al., <i>Cre recombinase-mediated site-specific recombination between plant chromosomes</i> , <i>Proc. Natl. Acad. Sci. USA</i> , 91:1706-1710 (1994).	
1,100,000		Sauer, Brian, <i>Manipulation of Transgenes by Site-Specific Recombination: Use of Cre Recombinase</i> , <i>Methods in Enzymology</i> , 225:890-900 (1993).	
1,100,000		Barinaga, Marcia, "Knockout mice: round two," <i>Science</i> , 265:26-28 (1994).	
1,100,000		Gu, Hui, Marth, James D., Orban, Paul C., Moosmann, Horst and Rajewsky, Klaus, "Deletion of a DNA of a polymerase II gene segment in T cells using cell type-specific gene targeting," <i>Science</i> , 265:103-106 (1994).	

Tech-Enabled New Data in the Loan Market

	Rationale	Sources	Users	
			Online	Banks
Cash Flow	- Best measure of firm fundamentals - Gauges ability to repay and manage repayment cycle	 	✓	
Credit Bureau	- Useful for evaluating consumer risk, but limited for business risk - Weak predictor of SMB creditworthiness	   	✓	✓
Firmographic	Risk correlated to revenue and profit level, size, age, industry, geography, customer size, owner education	   	✓	
Social	Volatile day to day, but viewed holistically can be a predictor of riskiness, esp. in retail and restaurants	   	✓	



Source: OnDeck.

Tech-Enabled Lending: Online versus Traditional

	 LendingClub	 SoFi	 PROSPER	 GreenSky	 CAN CAPITAL	 OnDeck	 Funding Circle	 Kabbage	 CommonBond
HQ	San Francisco, CA	San Francisco, CA	San Francisco, CA	Atlanta, GA	New York, NY	New York, NY	London, UK	Atlanta, GA	New York, NY
Founded	2007	2011	2005	2006	1998	2007	2009	2009	2011
Cumulative Originations ⁽¹⁾	~\$24.6BN	~\$16.0BN	~\$8.0BN	~\$7.0BN	~\$6.0BN	~\$6.0BN	~\$3.0BN	~\$2.5BN	~\$600MM
Model	Market Place Balance Sheet	Balance Sheet Market Place	Market Place	Bank Partner	Balance Sheet	Balance Sheet Market Place Bank Partner	Market Place Permanent Capital	Balance Sheet Market Place	Market Place
Products	- Personal Loans - Business Loans	- Student Loan Refi - Mortgage - Personal Loans - Investment Mgmt	- Personal Loans - Business Loans	- Home Improvement Loans - Purchase Finance	- Business Loans - Installment Loan - Merchant Cash Advance	- Business Loans - Lines of Credit	Business Loans	- Business Loans - Peer-to-Peer Loans - Lines of Credit	- Student Loans - Personal Loans
	Regulatory Filings indicate LC may start holding loans on balance sheet ⁽²⁾								

Source: SunTrust.

Balance Sheet versus Marketplace Lenders

Balance Sheet Lenders

Hold loans on their own balance sheet; bear all credit risk

Banks (traditional); **Fintech balance sheet lenders**: LendingClub (shifted to bank charter in 2021), Upstart (uses AI underwriting, sells loans to bank partners)

Upstart: 1,600+ variables vs. FICO's 22; claims 27% fewer defaults at same approval rate; approved 43% more Black and Hispanic borrowers than traditional models

Marketplace Lenders

Connect borrowers directly to institutional investors; platform does not hold credit risk

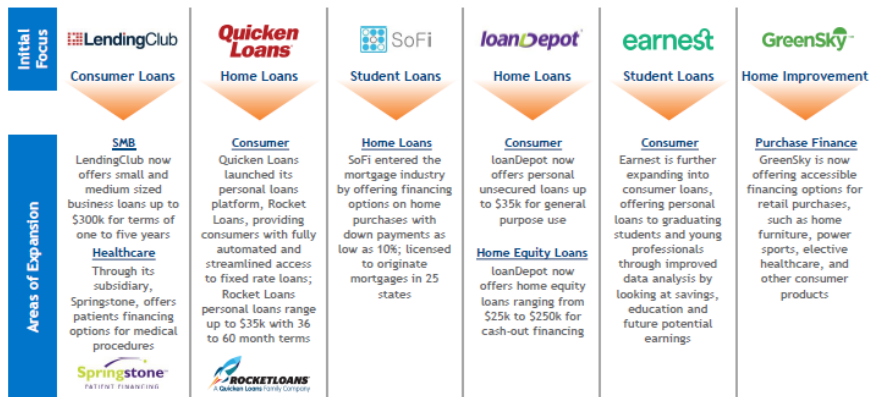
Examples: Prosper, Funding Circle (SME), SoFi (student/personal)

Marketplace lending holds **one-fifth of U.S. personal loan market share**; growing share of SME lending in the UK

Key Lesson

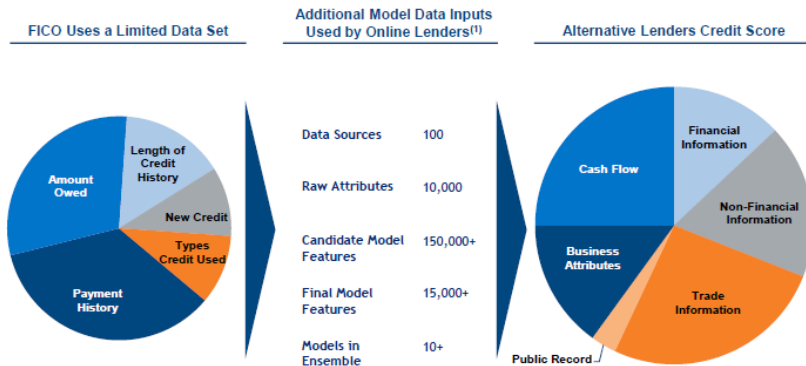
Tech-enabled lending democratizes access to credit — but regulatory arbitrage between banks and non-bank lenders creates risk that the 2008 crisis exposed in securitization markets.

Tech-Enabled Lending: Expanding into New Areas



Source: SunTrust.

Alternative Lenders: A Wealth of Data to Measure Risk



Source: OnDeck.

AI in Investment Decision-Making

McKinsey&Company

High Tech

Interview

McKinsey Quarterly

June 2017

A machine-learning approach to venture capital

In this interview, Hone Capital managing partner Veronica Wu describes how her team uses a data-analytics model to make better investment decisions in early-stage start-ups.

AI in Asset Management

Those big clients Ellis was initially worried about clearly like what they're seeing and have rushed to join Man's algorithm-centric funds. All told, four Man funds collectively managing \$12.3 billion are incorporating AI. Assets under management at Man have surged about 77 percent since the beginning of 2014. AHL Dimension fund assets have more than quintupled since then.

AI Skills for Finance Professionals

Since 2019, **CFA Institute** requires AI and data science literacy for the Level I and II exams — the exam now tests ML model evaluation, Python basics, and big data concepts

CPA (AICPA) integrating data analytics and AI tools into audit practice exams from 2024

FRM (GARP) added machine learning for risk management to the Part II curriculum

Career implication: finance professionals who cannot read a confusion matrix or interpret a random forest importance plot will be at a disadvantage

Unchecked algorithms
Machine-learning promises to shake up large swathes of finance

As banks from trading to credit assessment to fraud prevention, machine learning is advancing.



The
Economist

May 25th 2017

Machine Learning: Methods and Algorithms

Supervised versus Unsupervised Machine Learning

Supervised Learning

Learns from labeled training data to predict outcomes or classify observations

Classification: K-nearest neighbors, logistic regression, support vector machines, gradient boosting

Regression: linear, lasso (L1), ridge (L2), decision trees, random forests, XGBoost

Finance: credit default prediction, fraud detection, earnings surprise classification

Reinforcement Learning

Agent learns by interacting with an environment and maximizing cumulative reward

Applications: optimal execution, robo-advisory rebalancing, market-making algorithms

Unsupervised Learning

Finds hidden structure in unlabeled data

Clustering: K-means (segment customers), hierarchical clustering (group similar stocks)

Dimensionality reduction: PCA (risk factors from returns), LDA (topic modeling of 10-K filings)

Gu, Kelly, and Xiu (2020, *RFS*): ML dramatically outperforms linear factor models for cross-sectional stock returns

Key Lesson

Supervised learning predicts known outcomes; unsupervised learning discovers unknown patterns. Finance uses both — often in hybrid pipelines.

Evaluating Machine Learning Models

Training and Testing

Training sample: minimize in-sample prediction errors (RSS for regression, cross-entropy for classification)

Test sample: evaluate out-of-sample performance; never used during training

Validation sample: select among competing models (hyperparameter tuning)

Overfitting

Model fits training data perfectly but **fails on new data**

Classic symptom: training accuracy 99%, test accuracy 60%

Bias-variance tradeoff: complex models have low bias but high variance; simple models the reverse

Cross-Validation

k-fold: divide data into k folds; train on k-1, test on fold i; rotate until all folds used as test sets

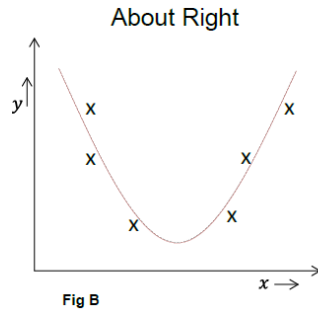
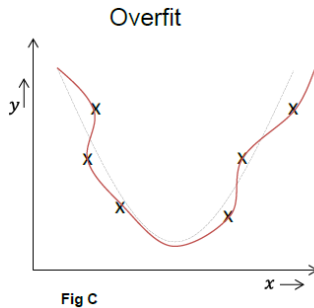
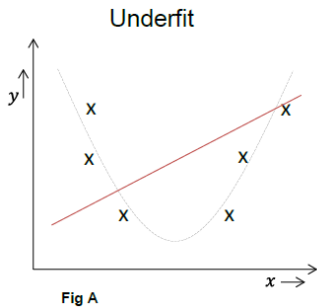
Aggregate prediction errors across all folds for robust model selection

Finance caution: use *time-series splits* (train on past, test on future) — standard k-fold can leak future information

Key Lesson

Data snooping bias is the finance version of overfitting — running enough specifications until finding a “significant” result is not evidence of a real effect.

Overfitting versus Underfitting



Source: J.P. Morgan.

Decision Trees: Predicting Baseball Salaries

Decision tree recursively splits the feature space to minimize prediction error (RSS for regression, Gini for classification)

Classic textbook example (ISLR): predict baseball player salary using Years and Hits; tree splits sample at Years < 4.5, then Hits < 117.5

Interpretable: the resulting tree can be displayed visually and explained to non-technical users — crucial for regulatory compliance

Finance use: simple decision trees are used in credit scorecards where regulators require explainable adverse action reasons



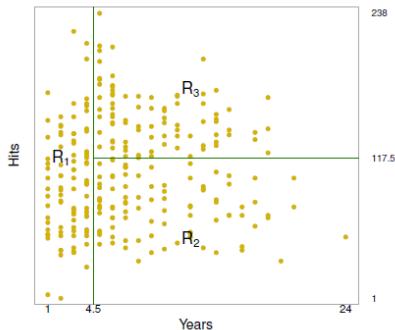
Decision Trees: Partition Regions

Each terminal node (leaf) of the tree corresponds to a region of the predictor space

R1 = {Years < 4.5}; **R2** = {Years ≥ 4.5, Hits < 117.5}; **R3** = {Years ≥ 4.5, Hits ≥ 117.5}

Predicted salary for each region: R1 = \$165,174;
R2 = \$402,834; R3 = \$845,346

Clear intuition: experienced players with high hit counts earn the most — exactly what domain knowledge would predict



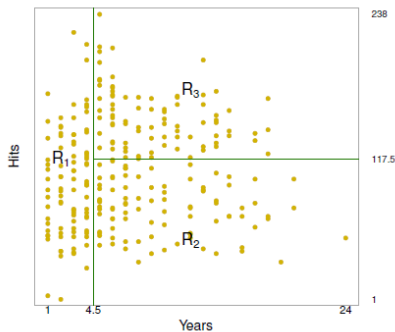
Decision Trees: 3D Partition Visualization

The 3D surface shows predicted log salary as a function of Years and Hits

Recursive binary splitting: at each step, choose the predictor and cutpoint that minimizes RSS across the resulting two regions

Stopping rule: continue splitting until no region contains more than a specified minimum number of observations (e.g., 5)

Pruning: grow a large tree then prune back using cost complexity to avoid overfitting — analogous to regularization in linear models



Building and Pruning Regression Trees

Recursive Binary Splitting: Step-by-Step

The Algorithm

At each step, find the predictor X_j and cutpoint s that minimizes: $\sum_{i: x_i \in R_1} (y_i - \hat{y}_{R_1})^2 + \sum_{i: x_i \in R_2} (y_i - \hat{y}_{R_2})^2$

Repeat: split one of the existing regions to further minimize RSS

Stop when a region has fewer than n_{min} observations (e.g., $n_{min} = 5$)

Greedy: makes the locally optimal split at each step — not guaranteed to be globally optimal

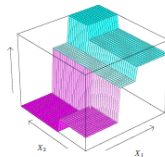
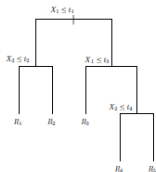
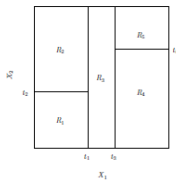
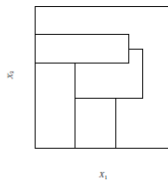
Cost Complexity Pruning

Grow a large tree T_0 ; then find the subtree $T \subset T_0$ minimizing: $\sum_{m=1}^{|T|} \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$

α is the penalty for tree size (analogous to LASSO penalty λ); chosen by cross-validation

Finance parallel: L1/L2 regularization in linear factor models serves the same bias-variance role

Recursive Binary Splitting: Visual Example

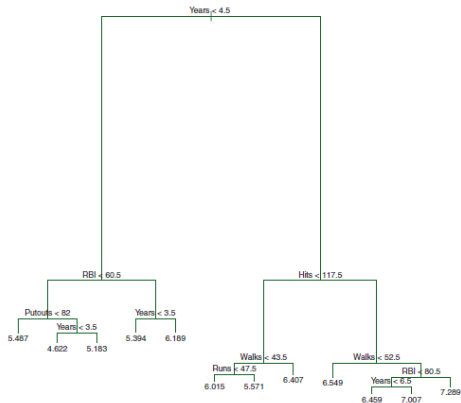


Building a Decision Tree

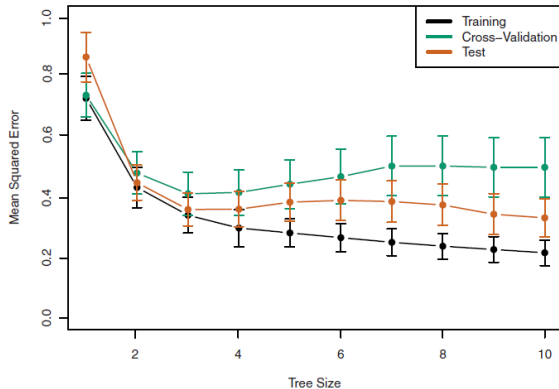
Algorithm 8.1 *Building a Regression Tree*

1. Use recursive binary splitting to grow a large tree on the training data, stopping only when each terminal node has fewer than some minimum number of observations.
 2. Apply cost complexity pruning to the large tree in order to obtain a sequence of best subtrees, as a function of α .
 3. Use K-fold cross-validation to choose α . That is, divide the training observations into K folds. For each $k = 1, \dots, K$:
 - (a) Repeat Steps 1 and 2 on all but the k th fold of the training data.
 - (b) Evaluate the mean squared prediction error on the data in the left-out k th fold, as a function of α .Average the results for each value of α , and pick α to minimize the average error.
 4. Return the subtree from Step 2 that corresponds to the chosen value of α .
-

An Unpruned Tree



Trees and Cross-Validation: Selecting Optimal Size



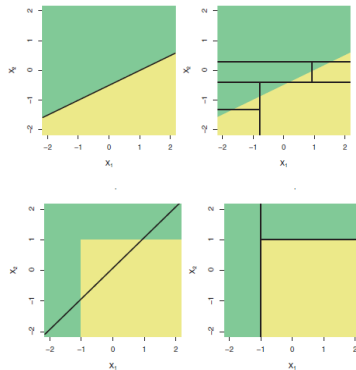
Linear Model versus Tree Model: When to Use Each

Linear model fits better when the true decision boundary is approximately linear (e.g., credit default risk as a function of debt-to-income ratio)

Tree model fits better when the boundary is complex and nonlinear (e.g., fraud detection with interaction effects between transaction time, location, and amount)

ISLR recommendation: no universally superior method; try both and use cross-validation to select

Finance practice: many quantitative funds combine linear factor models with tree-based ML in **hybrid factor models**



Ensemble Methods: Bagging, Random Forests, Boosting

Why Ensemble Methods?

Individual trees have **high variance**: small data changes produce very different trees

Ensemble methods **average many trees** to reduce variance while keeping low bias

Combine predictions of many weak learners into one strong learner

Bagging (Bootstrap Aggregation)

Draw B bootstrap samples; fit one tree per sample; **average predictions** (regression) or **majority vote** (classification)

Large B does not cause overfitting; choose B until OOB error stabilizes

OOB error: observations not in a bootstrap sample serve as a free validation set

Random Forests

Bagging with one twist: at each split, only a **random subset of m predictors** considered (typically $m = \sqrt{p}$)

Decorrelates trees: even if one strong predictor dominates, not all trees use it

Variable importance: RSS reduction per predictor across all trees — used to rank signals in quant models

Key Lesson

Random forests are the “workhorse” of applied ML in finance: robust, interpretable via variable importance, and hard to overfit with proper tuning.

Gradient Boosting: Sequential Tree Learning

How Boosting Differs from Bagging

Bagging builds trees **in parallel** on independent bootstrap samples; boosting builds trees **sequentially**

Each new tree fits the **residuals** (errors) of the ensemble so far — iteratively correcting mistakes

Gradient boosting: frames each iteration as gradient descent in function space; most popular implementation is XGBoost

XGBoost and LightGBM are the most-used algorithms in Kaggle competitions and quantitative finance since 2016

Key Hyperparameters

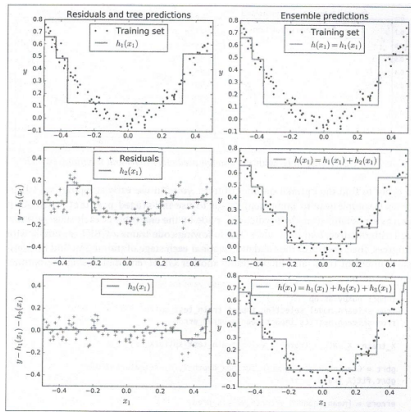
Number of trees B : more trees improve fit but risk overfitting; use early stopping

Learning rate λ : shrinks each tree's contribution; smaller λ requires more trees but generalizes better

Tree depth: shallow trees (“stumps”) for boosting vs. deeper trees for bagging

Finance application: XGBoost is widely used for consumer credit default prediction, replacing legacy logistic regression scorecards

Gradient Boosting: Visualization



Source: Geron (2017), *Hands-On Machine Learning with Scikit-Learn and TensorFlow*.

Gradient Boosting: Algorithm

Algorithm 8.2 *Boosting for Regression Trees*

1. Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all i in the training set.
2. For $b = 1, 2, \dots, B$, repeat:
 - (a) Fit a tree $\hat{f}^b$ with d splits ($d + 1$ terminal nodes) to the training data (X, r) .
 - (b) Update $\hat{f}$ by adding in a shrunk version of the new tree:

$$\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x). \quad (8.10)$$

- (c) Update the residuals,

$$r_i \leftarrow r_i - \lambda \hat{f}^b(x_i). \quad (8.11)$$

3. Output the boosted model,

$$\hat{f}(x) = \sum_{b=1}^B \lambda \hat{f}^b(x). \quad (8.12)$$

Gradient Boosting: Parameter Sensitivity

Aspect	Bagging	Random Forest	Gradient Boosting
Training	Parallel, independent trees	Parallel, independent trees with feature randomness	Sequential, each tree corrects previous errors
Focus	Reduce variance	Reduce variance further (decorrelation)	Reduce bias (and variance), improve accuracy
Tree depth	Usually deep (high-variance trees)	Deep	Shallow (weak learners)
Combination	Average / majority vote	Average / majority vote	Weighted sum (additive model)
Overfitting risk	Lower than single tree, but still possible	Lower (stronger decorrelation)	Higher risk if too many trees or too deep (needs tuning: learning rate, regularization)
Typical performance	Good	Better	Often best (state-of-the-art when tuned, e.g. XGBoost, LightGBM, CatBoost)

Support Vector Machines

Support Vector Machines: Overview

Background and Strengths

Developed in the 1990s (Vapnik); one of the best “out-of-the-box” classifiers, especially for small-to-medium datasets

Excels when features outnumber observations (e.g., text analysis with 50,000 word features and 500 documents)

Performs well in **image analysis** (early computer vision for document OCR) and **text classification** (spam detection, news sentiment)

Deep connections to logistic regression: both minimize similar loss functions

Key Idea: Separating Hyperplane

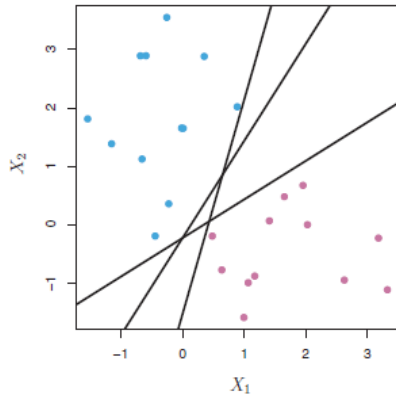
For a two-class problem in p -dimensional space, find a **hyperplane** that separates class +1 from class -1

In 2D: a line; in 3D: a plane; in high dimensions: the analog

The **maximal margin classifier** finds the hyperplane with the widest margin (distance to the nearest data points from either class)

Support vectors: only the observations closest to the margin determine the classifier — robust to other data points

Separating Hyperplanes



Maximal Margin Classifier: Intuition

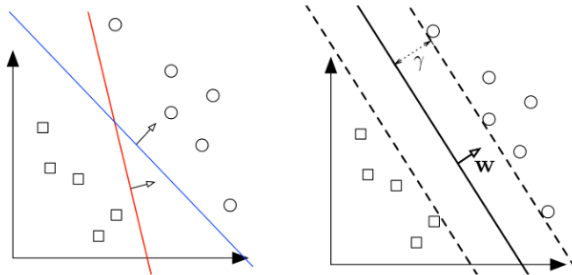


Figure 1: (Left:) Two different separating hyperplanes for the same data set. (Right:) The maximum margin hyperplane. The margin, γ , is the distance from the hyperplane (solid line) to the closest points in either class (which touch the parallel dotted lines).

SVM answer: the hyperplane that maximizes the margin to the nearest data points from both classes.

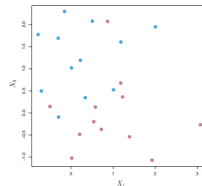
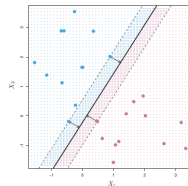
Maximal Margin Classifier: Properties and Limitations

Only support vectors matter: the classifier depends solely on the closest observations, not on all training data — a strength (robustness) and a weakness (sensitivity to outliers)

Sensitivity: adding one outlier near the margin can completely change the classifier

Non-separable data: if classes overlap, a perfect separating hyperplane does not exist — the maximal margin classifier fails

Solution: the **Support Vector Classifier** (soft margin) allows some misclassification



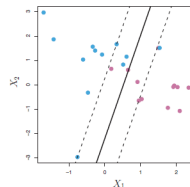
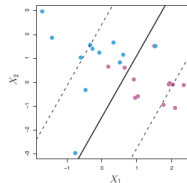
Support Vector Classifier

The **soft margin** SVC solves an optimization problem that allows some observations to be on the wrong side of the margin, controlled by tuning parameter C

Small C : narrow margin, few violations, low bias, high variance (risk of overfitting)

Large C : wide margin, more violations, higher bias, lower variance (more robust to individual observations)

C is chosen by cross-validation; analogous to regularization strength λ in ridge/lasso regression



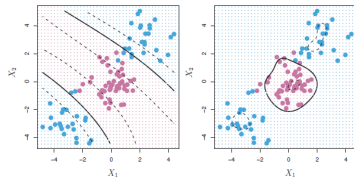
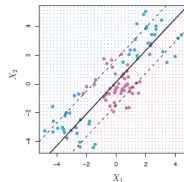
Support Vector Machines: Nonlinear Boundaries

The linear SVC fails when the true class boundary is nonlinear (e.g., circular or spiral patterns in the feature space)

Kernel trick: implicitly map data to a higher-dimensional space where classes become linearly separable, without computing the mapping explicitly

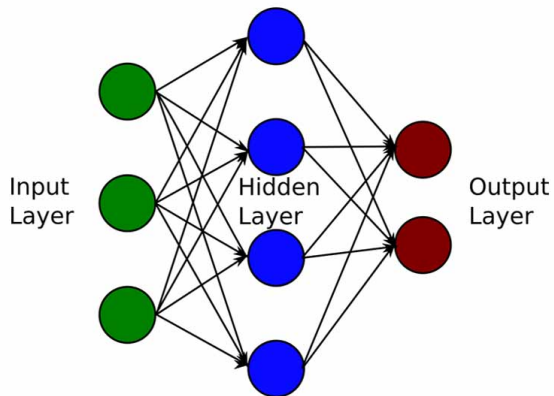
Polynomial kernel (degree d): captures polynomial interactions; **radial (RBF) kernel:** captures local neighborhoods; exponentially decaying influence

Finance: radial kernel SVM used for fraud detection where fraud patterns form irregular clusters in feature space



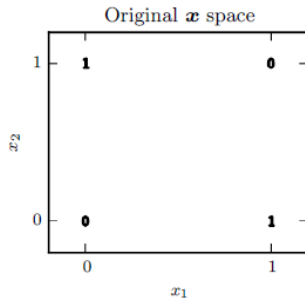
Deep Learning

Deep Learning: Feedforward Neural Networks



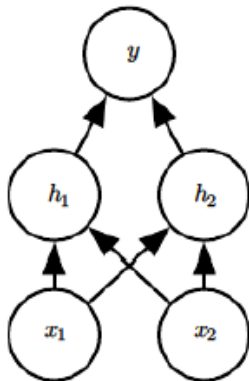
Source: Goodfellow, Bengio, and Courville (2016), *Deep Learning*, MIT Press.

Deep Learning: A Simple Neural Network Example

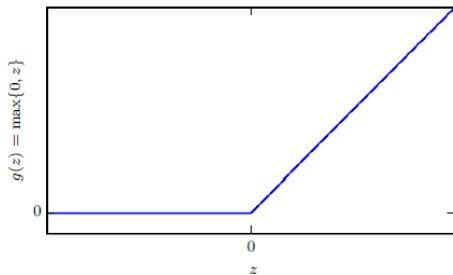


Source: Goodfellow et al. (2016)

Deep Learning: Forward Propagation

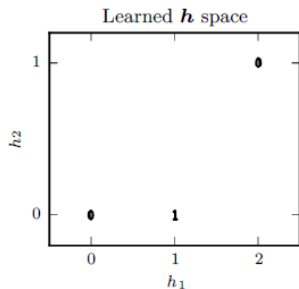


The ReLU Activation Function



Source: Goodfellow et al. (2016)

Deep Learning: Network with Multiple Layers



Source: Goodfellow et al. (2016)

Deep Learning Extensions in Finance

Key Architectures

Convolutional Neural Networks (CNN): process grid-structured data (images, charts); used for satellite imagery analysis in finance, earnings chart pattern recognition

Recurrent Neural Networks (RNN) and LSTM: process sequential data; predict time series (stock returns, default probabilities); capture long-range temporal dependencies

Transformers: “Attention Is All You Need” (Vaswani et al., 2017); replaced RNNs for NLP; powers GPT-4, Claude, BERT, BloombergGPT

Finance Applications

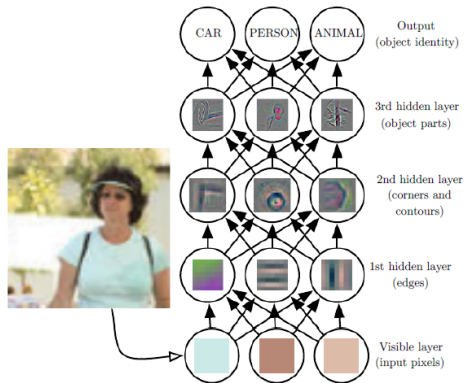
CNN: image-based analysis of satellite parking lots, shipping container counts, construction progress

LSTM: intraday price prediction, credit default probability sequences, earnings forecast revision chains

Transformer + finance: BloombergGPT (2023) trained on 363B financial tokens; outperforms general LLMs on financial NLP benchmarks (sentiment, NER, QA)

Graph Neural Networks: model counterparty networks for systemic risk; detect fraud rings in transaction graphs

Example of a Deep Learning Model



Source: Goodfellow et al. (2016).

Large Language Models (LLMs)

Large Language Models: Architecture and Training

What is an LLM?

A **large language model** predicts the next token given previous context; trained on hundreds of billions to trillions of tokens (web, books, code, papers)

Built on the **Transformer architecture** (Vaswani et al., 2017): multi-head self-attention, residual connections, layer normalization

Modern LLMs: GPT-4 (OpenAI), Claude 3.5 (Anthropic), Llama 3 (Meta), Gemini (Google)

Tokenization and Embeddings

Text split into **tokens** (subword units via BPE); each token mapped to a dense vector via embedding table

Positional encoding (sinusoidal or rotary) injects word-order information

Self-Attention Mechanism

Each token creates **query, key, value** vectors; relevance = dot product of query with keys, scaled and softmaxed into attention weights

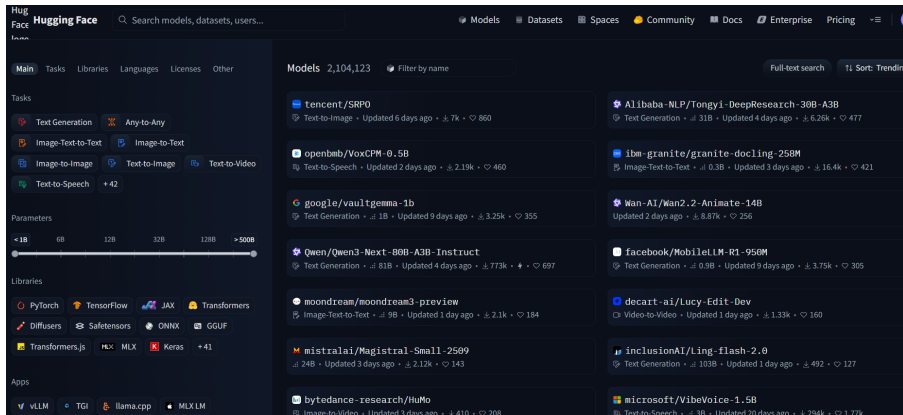
Multi-head attention allows simultaneous focus on different linguistic relationships

Finance: self-attention identifies which earlier earnings call statements are relevant for the current sentence

Key Lesson

LLMs are not “just” search engines — they generate novel, contextually appropriate text by learning the statistical structure of language at unprecedented scale.

Large Language Model Architecture



LLM Training: Pre-training and Fine-Tuning

Pre-Training

Objective: next-token prediction using cross-entropy loss across trillions of tokens

Data: web (Common Crawl), books, Wikipedia, code, scientific papers; cleaned and deduplicated

Scaling laws (Hoffmann et al., 2022): optimal training balances model size and data size; “compute-optimal” models train on more data rather than simply adding parameters (Chinchilla principle)

Training costs: GPT-4 estimated at \$50–100 million in compute; frontier models are a capital-intensive business

Fine-Tuning

Supervised fine-tuning (SFT): train on (prompt, ideal response) pairs to follow instructions

RLHF (Reinforcement Learning from Human Feedback): build a reward model from human preference rankings; update LLM to maximize reward using PPO

Domain adaptation: continue pre-training on financial text (SEC filings, Bloomberg news, earnings calls) to specialize; BloombergGPT used this approach

Parameter-efficient fine-tuning: LoRA adds trainable low-rank matrices without changing base model weights

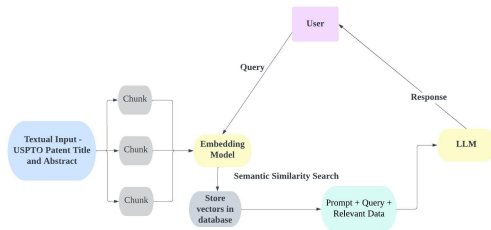
Retrieval-Augmented Generation (RAG)

LLMs have fixed context windows and knowledge cutoff dates — **RAG** solves this by retrieving relevant documents at query time

Architecture (Lewis et al., 2020): (1) embed the query; (2) retrieve top- k passages from a vector database; (3) concatenate retrieved passages with the query; (4) LLM generates a grounded response

Finance use: SEC filing Q&A (“What did Company X say about supply chain risk in its 2023 10-K?”); real-time regulatory guidance retrieval

Benefits: **reduced hallucination**, updated knowledge without retraining, and **cited sources** for regulatory explainability



LLM Inference: Decoding and Efficiency

Decoding Strategies

Greedy: always pick the most likely next token; fast but often repetitive

Beam search: maintain k candidate sequences simultaneously; better coherence, used in translation

Sampling with temperature: higher temperature T increases diversity; top- k restricts to k most likely tokens; top- p (nucleus sampling) keeps tokens whose cumulative probability reaches p

Finance: low temperature for factual SEC filing extraction; higher temperature for creative financial scenario analysis

Cost and Efficiency

Quantization: reduce weights from 32-bit float to 8-bit or 4-bit integers; 4× memory reduction with modest accuracy loss

Distillation: train a smaller “student” model to mimic a larger “teacher” model; Goldman Sachs runs distilled models for high-frequency compliance tasks

KV-cache: store computed attention keys and values to avoid recomputation during multi-turn conversations

Key Lesson

The cost of LLM inference is dropping rapidly — GPT-4-class intelligence now costs \$0.01 per 1,000 tokens, enabling financial AI applications at a scale impossible just two years ago.

Key Takeaways: Big Data, ML, and AI in Finance

What We Covered

Big Data (5Vs): structured and unstructured financial data from CRSP, Bloomberg, satellite imagery, credit card transactions, and social media

ML Fundamentals: supervised (decision trees, random forests, XGBoost, SVM), unsupervised (K-means, LDA, PCA), deep learning (CNN, LSTM, Transformer)

LLMs: transformer architecture, pre-training at scale, fine-tuning (SFT, RLHF), RAG; GPT-4, BloombergGPT, and Claude leading in finance applications

AI and Labor: augmenting vs. displacing AI; Chen and Wang (2025) show augmenting AI raises both TFP and firm value

Finance Applications and Risks

Credit: Upstart (1,600+ variables vs. FICO's 22); 27% fewer defaults; CFPB fair lending concerns

Trading: Renaissance Medallion (66% annual return 1988–2018); Gu, Kelly, Xiu (2020) show ML beats linear factor models

Operations: JPMorgan COiN (12,000 contracts in seconds); Klarna AI (700 agent equivalent); Morgan Stanley AI Assistant

Risks: overfitting, data snooping, algorithmic bias, flash crashes, black-box explainability under ECOA/EU AI Act